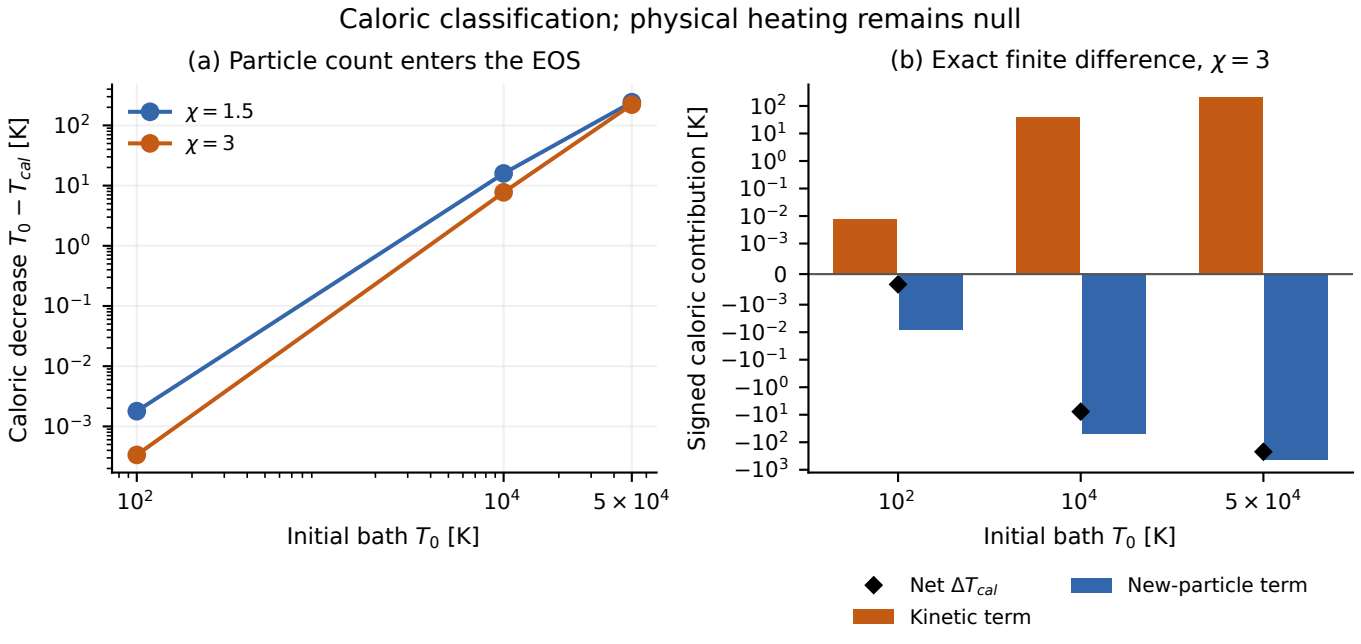
Caloric classification; physical heating remains null
(a) Particle count enters the EOS
$\chi = 1.5$
$\chi = 3$
Caloric decrease $T_0 - T_{cal}$ [K]
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{2}
10^{4}
5×10^{4}
Initial bath T_0 [K]
(b) Exact finite difference, $\chi = 3$
Signed caloric contribution [K]
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
-10^{-3}
-10^{-2}
-10^{-1}
-10^{0}
-10^{1}
-10^{2}
10^{2}
10^{4}
5×10^{4}
Initial bath T_0 [K]
Net ΔT_{cal}
New-particle term
Kinetic term